Quiz 4

Date: October 3 & 4, 2024

Name: ____

ERP ID: ____________________

Total Marks: 15 + 5 bonus points

Question 1 [5 points]

Construct a DFA using $\Sigma = \{a, b\}$ which accepts all strings that have an *a* at the second position and a *b* at the third last position.
(Construct a DFA, not define)

Solution

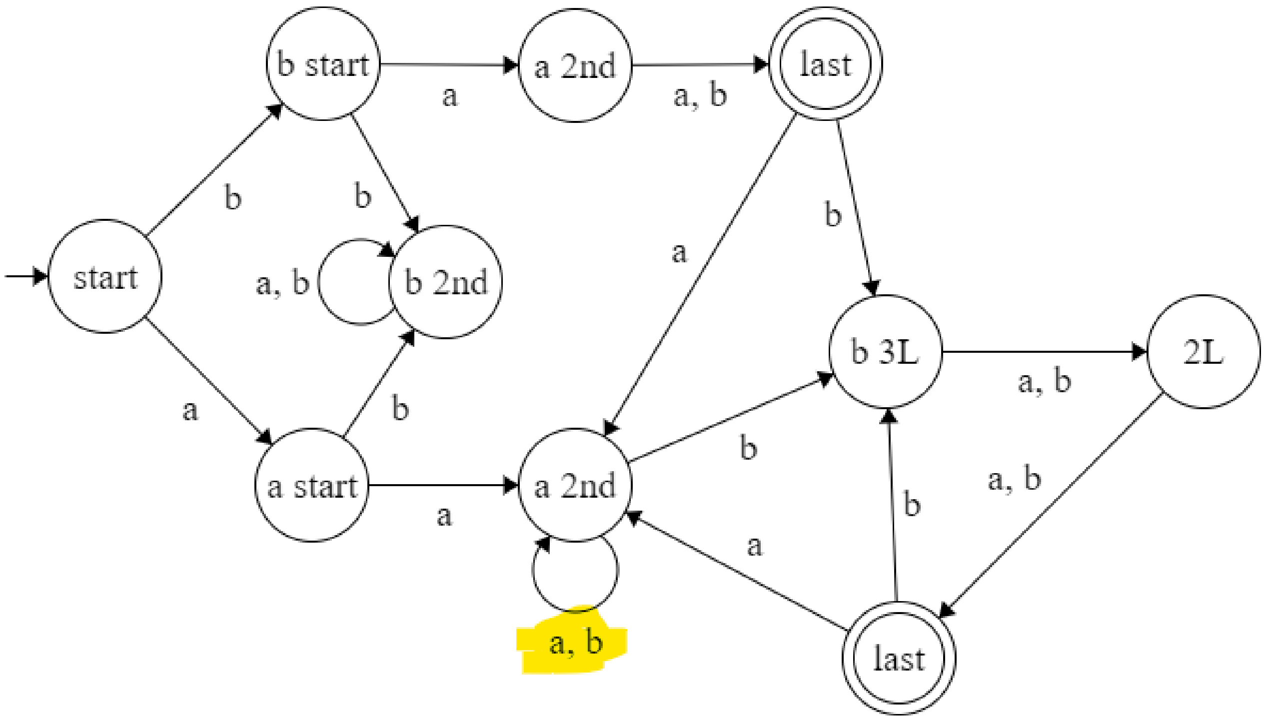
First identify the regular expression:

Figure (1): question 1 regular expression

$$(b\,a\,\Sigma) \cup (\Sigma\,a\,\Sigma^{*}\,b\,\Sigma\Sigma)$$

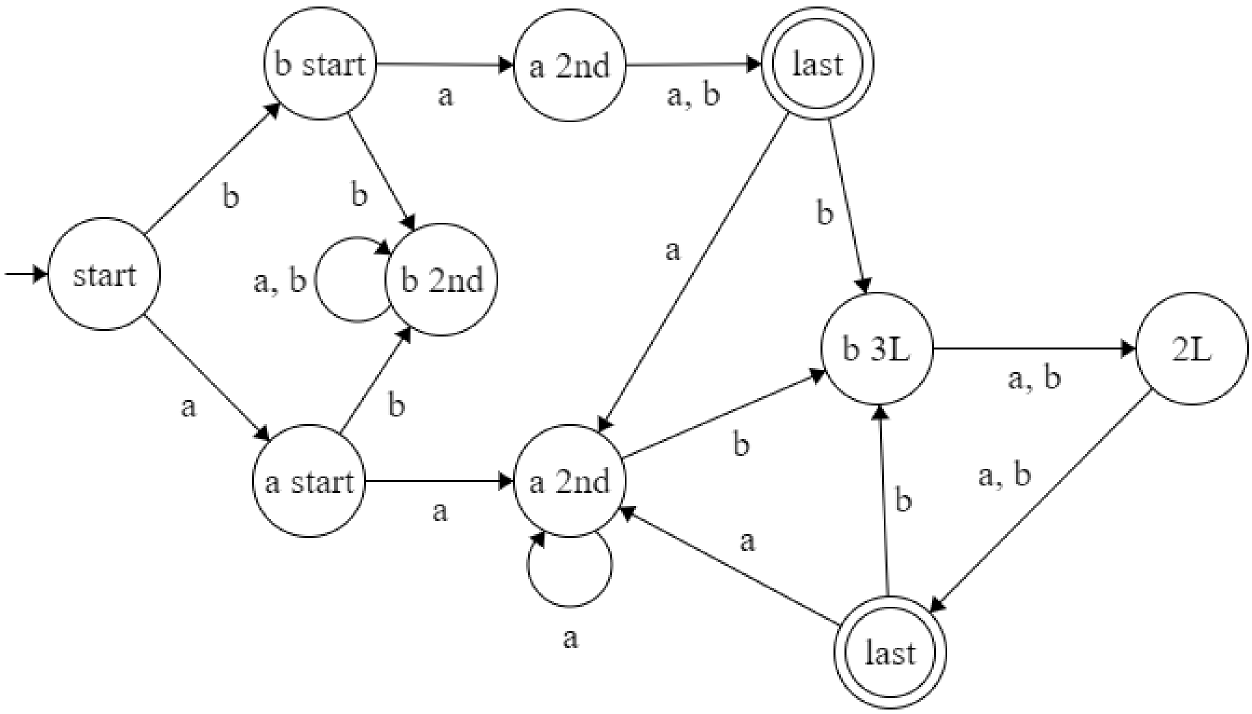
The NFA for this question would be:

Figure (2): question 1 NFA



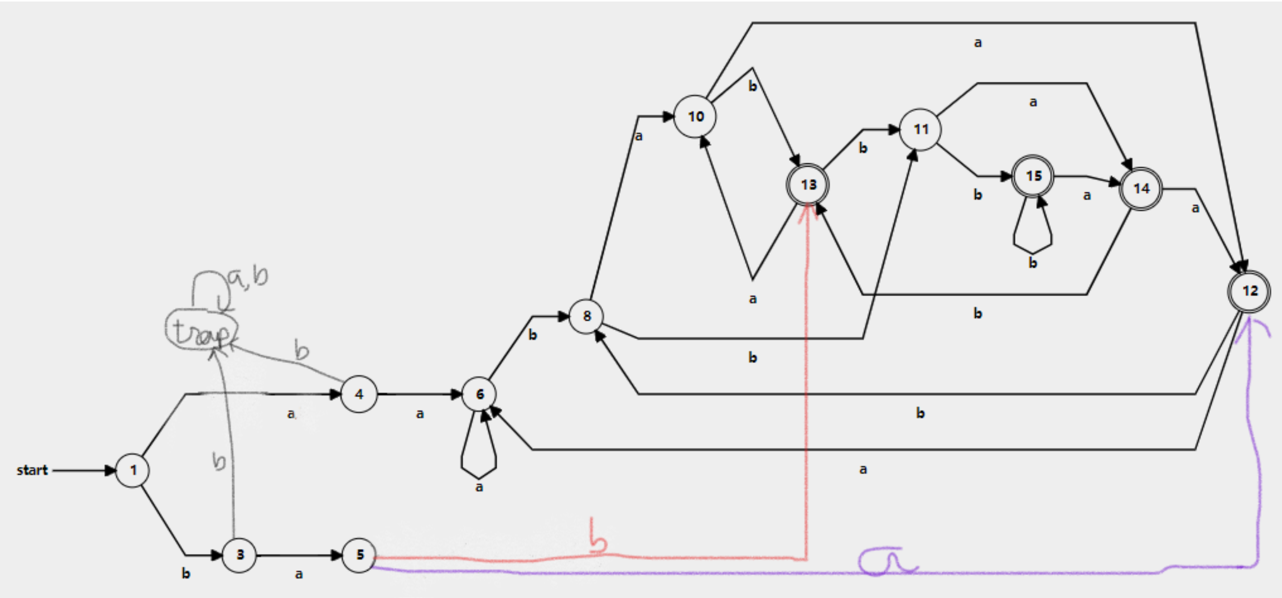
The acceptable DFA (this DFA is not fully correct but was accepted in the quiz):

Figure (3): question 1 acceptable DFA [2L means 2nd Last, 3L means 3rd last]



The full DFA for this language:

Figure (4): question 1 fully correct DFA



Marking Criteria

+ 1 for making a DFA
+ 1 for catering the $b.a.\Sigma$ string
+ 1 for adding transitions of a, b after the last $b.\Sigma.\Sigma$ (transitions for accepting states)
+ 1 for rejecting b at the second position
+ 1 for fully correct FA, no mistakes

if it is an NFA and it is fully functional and correct, error will be carried forward and only the first mark shall be lost.

Question 2 [5 points]

Prove that the following language $\{a^{n-2}b^nc^{n+2} : n \geq 2\}$ for $\Sigma = \{a, b, c\}$ is not regular.

Solution

To prove that this language is a non-regular language, we will perform a proof by contradiction. Assuming it is a regular language, we assume the pumping lemma with pumping length p is valid.

Assuming the string

$$s = a^p \; b^{p+2} \; c^{p+4}$$

this string s is in the language and $|s| = 3p + 6$ which is greater than the pumping length p, hence string s is valid. Condition $|s| \geq p$ is proved.

The three conditions of the pumping lemma are:
Condition 1: $xy^iz \in A$ where $i \geq 0$.
Condition 2: $|y| > 0$.
Condition 3: $|xy| \leq p$.

To split this string s into 3 parts x, y, z, we can perform the split,

$$x \to a^{p-1},\ y \to a,\ z \to b^{p+2}\, c^{p+4}$$

Condition 2, $|y| > 0$ is $|a| = 1 > 0$ thus validated.
Condition 3, $|xy| \leq p$ is $|a^{p-1}.a| = p \leq p$ thus validated.

Pumping down:
To perform condition 1, let us assume

$$i = 0 \text{ where the string will be } xz \text{ which is } a^{p-1}b^{p+2}c^{p+4}$$

The pumping lemma states that this pumped-down string should be in the language but after pumping down, we can see that this string is not in the language, as the number of a's should be 2 lesser than the number of b's but here they are 3 lesser hence property invalid.

Pumping up:
To perform condition 1, let us assume

$$i = 2 \text{ where the string will be } xyyz \text{ which is } a^{p-1}.a.a.b^{p+2}c^{p+4} \text{ which is } a^{p+1}b^{p+2}c^{p+4}$$

The pumping lemma states that this pumped-up string should be in the language but after pumping up, we can see that this string is not in the language, as now the number of a's is only 1 less than the number of b's when it should be 2 less hence property invalid.

Thus pumping lemma has been violated and therefore we have proof by contradiction that the language is not a regular language.

Marking Criteria

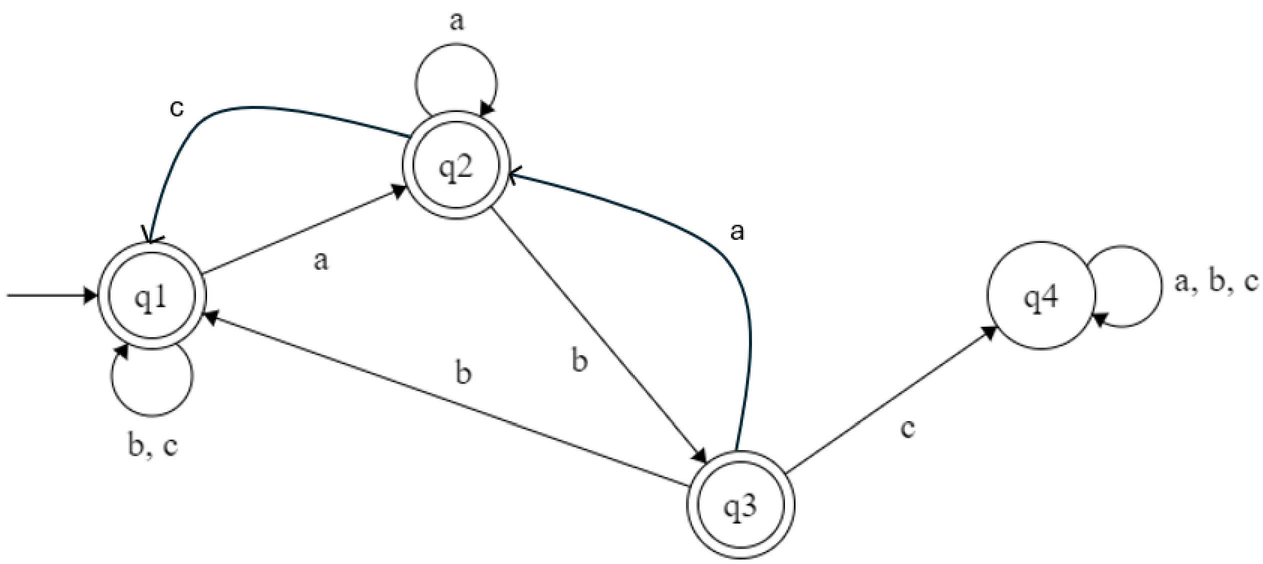
+ 1 for correctly defining a string s which is in the language and is in terms of p
+ 1 for validating condition 2 that $|y| > 0$
+ 1 for validating condition 3 that $|xy| \leq p$
+ 1 for correctly pumping up/down and showing the string after pumping
+ 1 for a concluding statement and stating that this pumped string is not in the language therefore this language is a non-regular language and a proof by contradiction

Question 3 [5 points]

Construct a DFA using $\Sigma = \{a, b, c\}$ that accepts any string that does not contain the substring "*abc*".

Solution

Figure (5): question 3 solution



Marking Criteria

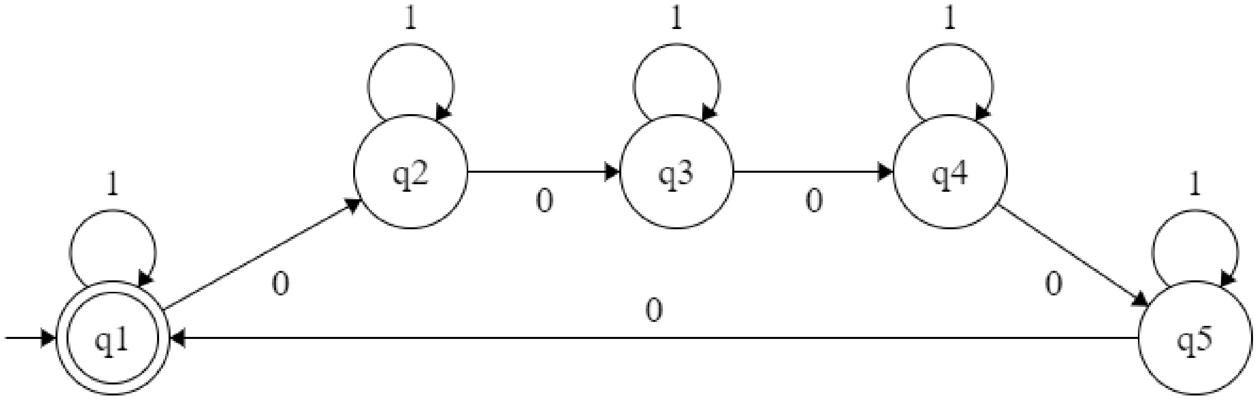
+ 1 for making a DFA
+ 1 for correct accepting states
+ 1 for accepting all non *abc* sub-strings
+ 1 for rejecting all *abc* sub-strings
+ 1 fully correct FA, no mistakes
if it is an NFA and it is fully functional and correct, error will be carried forward and only the first mark shall be lost.

Question 4: Bonus **[5 points]**

Construct a DFA of the language $\Sigma = \{0, 1\}$ which accepts all the strings with number of 0's in multiples of 5. (Construct a DFA, not define)

Solution

Figure (6): question 4 solution



Marking Criteria

+ 1 for making a DFA
+ 1 for accepting unlimited 1s (no check on 1s)
+ 1 for accepting zero 0s
+ 1 for counting 5 0s
+ 1 fully correct FA, no mistakes

if it is an NFA and it is fully functional and correct, error will be carried forward and only the first mark shall be lost.